

Solutions to MathPhys II colloquium 1

Problem 1. Infinite membrane: $\rho, h, \gamma, \sigma_e$

Potential: $U(x, y, z) = -U_0 \frac{z^2}{2}$

a) (General) wave equation describing small waves on a membrane subjected to external forces (from the lectures):

$$c^2 \nabla_{xy}^2 u + \frac{p}{\rho h} = u_{tt},$$

with $u = u(x, y, t)$, $c^2 = \frac{\gamma}{\rho h}$. We have:

$$\begin{aligned} p &= \frac{\vec{n} d\vec{F}}{dS} = \frac{dF_z}{dS} = \frac{d}{dS} (eE_z) \\ &= \left(\frac{d}{dS} e \right) \left(-\frac{\partial}{\partial z} \left(-U_0 \frac{z^2}{2} \right) \right) = \sigma_e U_0 z. \end{aligned}$$

So the **wave equation** becomes (z -position of the membrane is given by the displacement u):

$$\boxed{c^2 \nabla_{xy}^2 u + \frac{\sigma_e U_0}{\rho h} u = u_{tt}}. \quad (1)$$

To get the dispersion relation, we plug the plane wave:

$$u(\vec{x}, t) = e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

into this equation:

$$c^2 (-|k|^2) e^{i(\vec{k} \cdot \vec{x} - \omega t)} + \frac{\sigma_e U_0}{\rho h} e^{i(\vec{k} \cdot \vec{x} - \omega t)} = -\omega^2 e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

So the **dispersion relation** is:

$$\boxed{\left(\frac{\omega}{c} \right)^2 - |k|^2 = -\frac{\sigma_e U_0}{\rho h c^2}}. \quad (2)$$

b) From the derived relation and using the fact that ω doesn't change at domain edges, we see that the only effect of a change in the potential is a change of the magnitude of the wave vector $k := |k|$. We then have k_0 in the two outer regions and k_1 inside the strip. To find the relations between the amplitudes of transmitted and reflected waves, we use the ansatz:

$$u(\vec{x}, t) = \begin{cases} u_L; & x \leq -b \\ u_M; & -b \leq x \leq b, \\ u_R; & b \leq x \end{cases}$$

with:

$$\begin{aligned} u_L(\vec{x}, t) &= e^{i(k_0(x \cos(\alpha) + y \sin(\alpha)) - \omega t)} + R_1 e^{i(k_0(-x \cos(\beta) + y \sin(\beta)) - \omega t)}, \\ u_M(\vec{x}, t) &= T_2 e^{i(k_1(x \cos(\gamma) + y \sin(\gamma)) - \omega t)} + R_2 e^{i(k_1(-x \cos(\delta) + y \sin(\delta)) - \omega t)}, \\ u_R(\vec{x}, t) &= T_3 e^{i(k_0(x \cos(\phi) + y \sin(\phi)) - \omega t)}. \end{aligned}$$

This function needs to satisfy the boundary conditions:

1. Continuity at the domain edges (so that the membrane doesn't tear):

$$\text{a): } u_L|_{x=-b} = u_M|_{x=-b} \quad \text{and} \quad \text{b): } u_M|_{x=b} = u_R|_{x=b}.$$

2. Continuity of the normal derivative at the domain edges (not to have infinite acceleration at a domain edge):

$$\text{c): } \left. \frac{\partial}{\partial x} u_L \right|_{x=-b} = \left. \frac{\partial}{\partial x} u_M \right|_{x=-b} \quad \text{and} \quad \text{d): } \left. \frac{\partial}{\partial x} u_M \right|_{x=b} = \left. \frac{\partial}{\partial x} u_R \right|_{x=b}.$$

Plugging in the ansatz, we get:

1,a):

$$e^{i(k_0(-b \cos(\alpha) + y \sin(\alpha)) - \omega t)} + R_1 e^{i(k_0(b \cos(\beta) + y \sin(\beta)) - \omega t)} = T_2 e^{i(k_1(-b \cos(\gamma) + y \sin(\gamma)) - \omega t)} + R_2 e^{i(k_1(b \cos(\delta) + y \sin(\delta)) - \omega t)}.$$

Since this must hold for any y , that is, both sides have to oscillate equally with y , we must have:

$$k_0 \sin(\alpha) = k_0 \sin(\beta) = k_1 \sin(\gamma) = k_1 \sin(\delta).$$

A physical argument for this is that since the problem is translationally invariant in the y direction, the y component of the momentum is conserved. So we have:

$$\boxed{\alpha = \beta}, \quad \boxed{\gamma = \delta}$$

and

$$\boxed{\sin(\gamma) = \frac{k_0}{k_1} \sin(\alpha)}.$$

We're left with:

$$\boxed{e^{-ik_0 b \cos(\alpha)} + R_1 e^{ik_0 b \cos(\alpha)} = T_2 e^{-ik_1 b \cos(\gamma)} + R_2 e^{ik_1 b \cos(\gamma)}}. \quad (3)$$

1,b):

By the same procedure as above:

$$\sin(\phi) = \frac{k_1}{k_0} \sin(\gamma) = \sin(\alpha) \implies \boxed{\phi = \alpha}.$$

and

$$\boxed{T_2 e^{ik_1 b \cos(\gamma)} + R_2 e^{-ik_1 b \cos(\gamma)} = T_3 e^{ik_0 b \cos(\alpha)}}. \quad (4)$$

2,c):

$$\boxed{(ik_0) \cos(\alpha) e^{-ik_0 b \cos(\alpha)} - R_1 (ik_0) \cos(\alpha) e^{ik_0 b \cos(\alpha)} = T_2 (ik_1) \cos(\gamma) e^{-ik_1 b \cos(\gamma)} - R_2 (ik_1) \cos(\gamma) e^{ik_1 b \cos(\gamma)}}. \quad (5)$$

2,d):

$$\boxed{T_2 (ik_1) \cos(\gamma) e^{ik_1 b \cos(\gamma)} - R_2 (ik_1) \cos(\gamma) e^{-ik_1 b \cos(\gamma)} = T_3 (ik_0) \cos(\alpha) e^{ik_0 b \cos(\alpha)}}. \quad (6)$$

Equations (3-6) constitute the required **set of linear equations**. Solving it was not required in the colloquium. It can be done quickly with Mathematica using LinearSolve.

Bonus questions: Choose $U_0 < 0$ (the minus sign in the definition of the potential in the text of the problem was a lapsus which, however, doesn't affect the solution procedure of a) and b)). Then, the wave equation (1) is equivalent to the famous (classical version of) **Klein-Gordon equation**:

$$\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \psi - \nabla^2 \psi + \frac{m^2 c^2}{\hbar^2} \psi = 0$$

which is often given by its Lagrangian:

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \psi \partial_\nu \psi - \frac{1}{2} m^2 \psi^2.$$

It is (in the quantum version; quantum field theory) a generalization of the Schrödinger equation describing the dynamics of spinless relativistic particles with mass m . You will learn all about it at some of the courses on particle physics, advanced QM or quantum field theory.

The fact that the equation is describing relativistic particles could be more easily recognized from the dispersion relation (2), which is equivalent to the **relativistic energy-momentum relation**:

$$-p_\mu p^\mu = \left(\frac{E}{c} \right)^2 - P^2 = (mc)^2$$

or for the wave:

$$-k_\mu k^\mu = \left(\frac{\omega}{c} \right)^2 - k^2 = (mc)^2.$$

You already know these relations from the courses on modern physics.

The b) part of the problem can also be given a physical “motivation”, coming from solid-state physics, where electrons in external potential sometimes get a larger effective mass. Although, in solid-state physics the particles are never (or almost never) relativistic.

Problem 2. We are solving the diffusion equation (with no sources):

$$D \nabla^2 T = \frac{\partial T}{\partial t}, \quad (7)$$

where $D = \frac{\lambda}{\rho c_p}$. We're solving it on the domain $(x, y, z) \in [0, b] \times [0, b] \times [0, d]$, $t \in [0, \infty)$. The crucial point of the problem are the boundary conditions. These are the following:

- At the side planes that we are cooling, the temperature is fixed: $T|_{(0,y,z)} = T|_{(b,y,z)} = T_0$.
- At the planes, where the system is isolated, the (normal) current $\vec{j} \cdot \vec{n} = -\lambda \frac{\partial T}{\partial \vec{x}} \cdot \vec{n}$, thus the normal derivative, is fixed to 0:

$$\left. \frac{\partial T}{\partial y} \right|_{(x,0,z)} = \left. \frac{\partial T}{\partial y} \right|_{(x,b,z)} = 0, \quad (8)$$

$$\left. \frac{\partial T}{\partial z} \right|_{(x,y,0)} = 0. \quad (9)$$

- At the upper plane, where the chip is placed, we have a nonhomogeneous boundary condition for the current (normal derivative) - the current just below the chip is nonzero which is the consequence of the heat emitted by the chip:

$$\left. \frac{\partial T}{\partial z} \right|_{(x,y,d)} = -\frac{j_z}{\lambda} = \begin{cases} \frac{P}{\lambda a^2}; & (x, y) \in \left[\frac{b-a}{2}, \frac{b-a}{2} + a\right] \times \left[\frac{b-a}{2}, \frac{b-a}{2} + a\right] = f(x, y), \\ 0; & \text{elsewhere} \end{cases} \quad (10)$$

where we have used $j_z = -\frac{P}{S}$ below the chip.

To solve the equation, we first subtract the constant from the coordinate-dependent part of temperature:

$$T(x, y, z, t) = \tilde{T}(x, y, z, t) + T_0,$$

so that the boundary conditions at the cooled planes simplify:

$$\tilde{T}|_{(0,y,z)} = \tilde{T}|_{(b,y,z)} = 0. \quad (11)$$

To find $\tilde{T}$, we use the separation of variables ansatz:

$$\tilde{T}(x, y, z, t) = X(x) Y(y) Z(z) \tau(t) = R(x, y, z) \tau(t).$$

Plugging it into the diffusion equation and dividing by $\tilde{T}(x, y, z, t)$, we get:

$$\frac{\nabla^2 R}{R} = \frac{1}{D} \frac{\tau_t}{\tau} = \lambda.$$

We have used the standard argument: the left side only depends on the spatial coordinates, the right only on time, so they can only be equal if they are both equal to a constant λ . We have to distinguish three cases: $\lambda > 0$, $\lambda = 0$ and $\lambda < 0$. We see immediately that if $\lambda > 0$, τ is going to grow exponentially with time which is not physical. So we're left with two cases:

a) $\lambda = 0$. In this case, $\tau(t) = \text{const}$ so we get the **stationary solution**. We're solving the (3D) Laplace equation for the stationary profile:

$$\nabla^2 R_S(x, y, z) = 0.$$

By the standard procedure:

$$\frac{X_S''}{X_S} + \frac{Y_S''}{Y_S} + \frac{Z_S''}{Z_S} = 0$$

or:

$$\frac{X_S''}{X_S} + \frac{Y_S''}{Y_S} = -\frac{Z_S''}{Z_S} = -k_x^2 - k_y^2.$$

The index S denotes that the solution is stationary. The solutions to this equations are trigonometric functions in x and y directions and hyperbolic functions in z direction. Notice that the equation itself would in principle allow us to choose the hyperbolic solution in any one of the three directions. However, we need to satisfy the homogeneous (and equal to zero) boundary conditions in x and y directions, so we must choose z direction to carry the hyperbolic solution. We get in x direction:

$$X(x) = A_{n_x} \sin(k_{n_x} x) + B_{n_x} \cos(k_{n_x} x).$$

To satisfy the boundary conditions (11), $B_{n_x} = 0$ and $k_{n_x} = \frac{n_x \pi}{b}$, for $n_x = 1, 2, \dots$.

In y direction:

$$Y(y) = C_{n_y} \sin(k_{n_y} y) + D_{n_y} \cos(k_{n_y} y).$$

To satisfy the BCs (8), $C_{n_y} = 0$ and $k_{n_y} = \frac{n_y \pi}{b}$, for $n_y = 0, 1, 2, \dots$.

In z direction:

$$Z(z) = E_{n_x n_y} \sinh(k_{n_x n_y} z) + F_{n_x n_y} \cosh(k_{n_x n_y} z),$$

where $k_{n_x n_y} = \sqrt{k_{n_x}^2 + k_{n_y}^2}$. To satisfy the BC (9), $E_{n_x n_y} = 0$. Finally, we have to take care of the nonhomogeneous BC at the top of the cooling pad (10):

$$\left. \frac{\partial R_S}{\partial z} \right|_{(x,y,d)} = \sum_{n_x, n_y} A_{n_x n_y} \sin(k_{n_x} x) \cos(k_{n_y} y) k_{n_x n_y} \sinh(k_{n_x n_y} d) = f(x, y).$$

To find the constants $A_{n_x n_y}$, we multiply both sides by $\sin(k_{m_x} x) \cos(k_{m_y} y)$ and integrate over x and y ($\int_0^b \int_0^b dx dy$).

For $m_y \geq 1$:

$$\begin{aligned}
\sum_{n_x, n_y \in \mathbb{N}} \frac{b}{2} \delta_{n_x m_x} \frac{b}{2} \delta_{n_y m_y} A_{n_x n_y} k_{n_x n_y} \sinh(k_{n_x n_y} d) &= \left(\frac{b}{2}\right)^2 A_{m_x m_y} k_{m_x m_y} \sinh(k_{m_x m_y} d) \\
&= \int_0^b \int_0^b dx dy f(x, y) \sin(k_{m_x} x) \cos(k_{m_y} y) \\
&= \frac{P}{\lambda a^2} \int_{\frac{b-a}{2}}^{\frac{b+a}{2}} \int_{\frac{b-a}{2}}^{\frac{b+a}{2}} dx dy \sin(k_{m_x} x) \cos(k_{m_y} y) \\
&= \frac{-1}{k_{m_x} k_{m_y}} \frac{P}{\lambda a^2} \left[\cos\left(k_{m_x} \frac{b+a}{2}\right) - \cos\left(k_{m_x} \frac{b-a}{2}\right) \right] \\
&\quad \cdot \left[\sin\left(k_{m_y} \frac{b+a}{2}\right) - \sin\left(k_{m_y} \frac{b-a}{2}\right) \right] \\
&= \frac{4}{k_{m_x} k_{m_y}} \frac{P}{\lambda a^2} \sin\left(k_{m_x} \frac{b}{2}\right) \sin\left(k_{m_y} \frac{a}{2}\right) \\
&\quad \cdot \cos\left(k_{m_y} \frac{b}{2}\right) \sin\left(k_{m_y} \frac{a}{2}\right) \\
&= \begin{cases} \frac{4}{k_{m_x} k_{m_y}} \frac{P}{\lambda a^2} (-1)^{\frac{m_x-1}{2}} (-1)^{\frac{m_y}{2}} \sin\left(k_{m_x} \frac{a}{2}\right) \sin\left(k_{m_y} \frac{a}{2}\right); & \text{i)} \\ 0; & \text{ii)} \end{cases}
\end{aligned}$$

where the cases are: i) m_x odd $\wedge$ m_y even, ii) otherwise. To simplify the result we have used a couple of trigonometric identities.

For $m_y = 0$:

$$\begin{aligned}
\frac{b}{2} A_{m_x 0} k_{m_x 0} \sinh(k_{m_x 0} d) &= \frac{P}{\lambda a^2} \int_{\frac{b-a}{2}}^{\frac{b+a}{2}} \int_{\frac{b-a}{2}}^{\frac{b+a}{2}} dx dy \sin(k_{m_x} x) \\
&= \frac{-1}{k_{m_x}} \frac{P}{\lambda a^2} \left[\cos\left(k_{m_x} \frac{b+a}{2}\right) - \cos\left(k_{m_x} \frac{b-a}{2}\right) \right] a \\
&= \frac{2}{k_{m_x}} \frac{P}{\lambda a} \sin\left(k_{m_x} \frac{b}{2}\right) \sin\left(k_{m_x} \frac{a}{2}\right) \\
&= \begin{cases} \frac{2}{k_{m_x}} \frac{P}{\lambda a} (-1)^{\frac{m_x-1}{2}} \sin\left(k_{m_x} \frac{a}{2}\right); & m_x \text{ odd} \\ 0; & \text{otherwise} \end{cases},
\end{aligned}$$

which equals to the limit $m_y \rightarrow 0$ of the above formula with the difference of factor 2 on the left side.

So then, writing the result in a compact way:

$$A_{m_x m_y} = \begin{cases} \frac{(-1)^{\frac{m_x-1}{2}} (-1)^{\frac{m_y}{2}}}{k_{m_x} k_{m_y} k_{m_x m_y}} \frac{16P}{2^{\delta_{m_y,0}} \lambda a^2 b^2} \frac{\sin(k_{m_x} \frac{a}{2}) \sin(k_{m_y} \frac{a}{2})}{\sinh(k_{m_x m_y} d)}; & m_x \text{ odd} \wedge m_y \text{ even} \\ 0; & \text{otherwise} \end{cases}$$

and we get the stationary profile:

$$T_S(x, y, z) = \sum_{n_x, n_y} A_{n_x n_y} \sin(k_{n_x} x) \cos(k_{n_y} y) \cosh(k_{n_x n_y} z) + T_0.$$

b) $\lambda < 0$. We are solving for the **dynamical part** of the solution:

$$\frac{X_D''}{X_D} + \frac{Y_D''}{Y_D} + \frac{Z_D''}{Z_D} = \frac{1}{D} \frac{\tau_t}{\tau} = -|\lambda| = -k_{n_x}^2 - k_{n_y}^2 - k_{n_z}^2.$$

The boundary conditions that the dynamical part $\widetilde{T}_D(x, y, z, t) = X_D(x) Y_D(y) Z_D(z) \tau(t)$ has to satisfy are (8),(9),(11), but instead of (10) we have:

$$\left. \frac{\partial \widetilde{T}_D}{\partial z} \right|_{(x,y,d)} = 0,$$

which is equivalent to the isolated pad (without the chip). This is the case because the boundary condition (10) is nonzero, nonhomogeneous and time independent, so it would be impossible to satisfy it with a linear combination (with constant coefficients) of time dependent functions. In particular, (10) has been already taken care of with the stationary part of the solution. So, the full solution will satisfy the full boundary conditions (including the heath flow from the chip) even though the dynamical part corresponds to zero current at the upper plane of the pad.

We get for the **dynamical part** of the solution:

$$\boxed{\widetilde{T}_D(x, y, z, t) = \sum_{n_x, n_y, n_z \in \mathbb{N}} B_{n_x n_y n_z} \sin(k_{n_x} x) \cos(k_{n_y} y) \cos(k_{n_z} z) e^{-D \sqrt{k_{n_x}^2 + k_{n_y}^2 + k_{n_z}^2} t}},$$

where k_{n_x} and k_{n_y} are the same as above and $\boxed{k_{n_z} = \frac{n_z \pi}{d}, \text{ for } n_z = 0, 1, 2, \dots}$. Then the **full solution** is:

$$\begin{aligned} \boxed{T(x, y, z, t)} &= T_S(x, y, z) + \widetilde{T}_D(x, y, z, t) \\ &= \boxed{\sum_{n_x, n_y} \sin(k_{n_x} x) \cos(k_{n_y} y) [A_{n_x n_y} \cosh(k_{n_x n_y} z) +} \\ &\quad \boxed{+ \sum_{n_z} B_{n_x n_y n_z} \cos(k_{n_z} z) e^{-D \sqrt{k_{n_x}^2 + k_{n_y}^2 + k_{n_z}^2} t}]} + T_0 \end{aligned} \quad (12)$$

In order to get the constants $B_{n_x n_y n_z}$, we have to satisfy the initial condition:

$$T(x, y, z, 0) = T_0,$$

that is:

$$\sum_{n_z} B_{n_x n_y n_z} \cos(k_{n_z} z) = -A_{n_x n_y} \cosh(k_{n_x n_y} z).$$

Similarly as earlier, we multiply both sides by $\cos(k_{m_z} z)$ and integrate $\int_0^d dz$:

$$\begin{aligned} 2^{\delta_{m_z, 0}} \frac{d}{2} B_{n_x n_y m_z} &= -A_{n_x n_y} \int_0^d dz \cosh(k_{n_x n_y} z) \cos(k_{m_z} z) \\ &= -A_{n_x n_y} \frac{k_{n_x n_y} (-1)^{m_z} \sinh(k_{n_x n_y} d)}{k_{m_z}^2 + k_{n_x n_y}^2}, \end{aligned}$$

where from the first to the second line we have integrated by parts (twice). So, we get:

$$B_{n_x n_y m_z} = \frac{-2A_{n_x n_y}}{2^{\delta_{m_z,0}} d} \frac{k_{n_x n_y} (-1)^{m_z} \sinh(k_{n_x n_y} d)}{k_{m_z}^2 + k_{n_x n_y}^2}.$$

The dynamics after a long time will be dominated with the slowest decaying mode e^{-t/τ_τ} . This corresponds to the mode with the largest **time constant** $\tau_\tau = \frac{1}{D\sqrt{k_{n_x}^2 + k_{n_y}^2 + k_{n_z}^2}}$. We get it for $n_x = 1, n_y = n_z = 0$:

$$\tau_{\tau\max} = \frac{b}{D\pi}.$$

The **difference of the temperatures** on the top and the bottom of the pad just under the center of the chip is:

$$\begin{aligned} \boxed{\Delta T(t)} &= T\left(\frac{b}{2}, \frac{b}{2}, d, t\right) - T\left(\frac{b}{2}, \frac{b}{2}, 0, t\right) \\ &= \boxed{\sum_{\substack{m \in \mathbb{N} \\ n \in \mathbb{N} \cup \{0\}}} (-1)^{m-1} (-1)^n A_{(2m-1)_x (2n)_y} \left[\left(\cosh(k_{(2m-1)_x (2n)_y} d) - 1 \right) - \right.} \\ &\quad \left. - \sum_{j \in \mathbb{N} \cup \{0\}} \frac{4k_{(2m-1)_x (2n)_y} \sinh(k_{(2m-1)_x (2n)_y} d)}{2^{\delta_{j,0}} d \left((k_{(2j-1)_z})^2 + k_{(2m-1)_x (2n)_y}^2 \right)} e^{-D\sqrt{(k_{(2m-1)_x})^2 + (k_{(2n)_y})^2 + (k_{(2j-1)_z})^2} t} \right]. \end{aligned}$$

To get the last result, check for which n_x, n_y and n_z the expressions $\sin(k_{n_x} \frac{b}{2})$, $\cos(k_{n_x} \frac{b}{2})$ and $\cos(k_{n_z} d)$ give 1, -1 or 0 and take just nonzero terms.

Note: Problem 2 could equivalently be solved by solving the diffusion equation with sources:

$$\frac{\partial T}{\partial t} = D\nabla^2 T + \frac{q}{\rho c_p}$$

and taking:

$$q(x, y, z) = \frac{P}{\lambda a^2} \left[\Theta\left(x - \frac{b-a}{2}\right) - \Theta\left(x - \frac{b+a}{2}\right) \right] \left[\Theta\left(y - \frac{b-a}{2}\right) - \Theta\left(y - \frac{b+a}{2}\right) \right] \delta(z-d).$$

In this case the correct boundary conditions would be homogeneous, for the isolated pad:

•

$$T|_{(0,y,z)} = T|_{(b,y,z)} = T_0,$$

•

$$\frac{\partial T}{\partial y} \Big|_{(x,0,z)} = \frac{\partial T}{\partial y} \Big|_{(x,b,z)} = \frac{\partial T}{\partial z} \Big|_{(x,y,0)} = \frac{\partial T}{\partial z} \Big|_{(x,y,d)} = 0.$$

The solution (also the time-dependent) could be obtained by projecting to the eigenfunctions of the Laplace operator (and solving nonhomogeneous ODEs for the time-dependent case). You can check that the results match the ones above.